To say the same thing more directly, we consider a small region in flat spacetime. It has some entropy, because the modes inside the region are entangled with those outside. Now imagine changing the quantum state a little bit, so that we decrease the amount by which that region is entangled, and therefore decrease its entropy. In Jacobson's picture, the area bounding our region changes in response, shrinking by a bit. And he shows that this response of the geometry of spacetime to a change in the quantum state is equivalent to Einstein's equation of general relativity, relating curvature to energy.

This was the beginning of a surge of interest in what is now called "entropic" or "thermodynamic" gravity; other important contributions were made by Thanu Padmanabhan (2009) and Erik Verlinde (2010). The behavior of spacetime in general relativity can be thought of as simply the natural tendency of systems to move toward configurations of higher entropy.

This is a fairly radical change of perspective. Einstein thought in terms of energy, a definite quantity associated with particular configurations of stuff in the universe. Jacobson and others have argued that we can reach the same conclusions by thinking about entropy, a collective phenomenon that emerges from the mutual interaction of many small constituents of a system. This simple shift in focus might offer a crucial way forward in our quest to discover a fundamentally quantum theory of gravity.

Jacobson wasn't himself proposing a theory of quantum gravity; he was pointing to a new way to derive Einstein's equation for classical general relativity, with quantum fields acting as the source of energy. The appearance of words like "area" and "region of space" should indicate to us that the above discussion treated spacetime as a tangible, classical thing. But given the central role that entanglement entropy plays in his derivation, it's natural to ask whether we might adapt the basic ideas to an approach that is more intrinsically quantum from the start, where space itself emerges from the wave function.

In Many-Worlds, a wave function is just an abstract vector living within the super-high-dimensional mathematical construct of Hilbert space.